Exercise 1.15B

1 Evaluate each of the following.

a $\frac{6}{\sqrt{2}} \times \frac{7}{\sqrt{2}}$

b $\frac{21}{\sqrt{3}} \times \frac{4}{\sqrt{3}}$

c $\frac{6}{\sqrt{5}} \times \frac{15}{2\sqrt{5}}$

d $\frac{5}{3\sqrt{8}} \times \frac{4}{8\sqrt{2}}$

2 Write each of the following with a rational denominator.

a $\frac{8}{\sqrt{2}}$

b $\frac{21}{\sqrt{3}}$

c $\frac{40}{\sqrt{5}}$

d $\frac{3}{\sqrt{2}}$

e $\frac{255}{\sqrt{10}}$

f $\frac{15}{3\sqrt{3}}$

g $\frac{144}{5\sqrt{27}}$

3 a Find the area of a square that has a perimeter of $28\sqrt{5}$ cm.

b Find the area of a rectangle with sides that are $2\sqrt{10}$ cm and $8\sqrt{10}$ cm long.

c Find the area of a triangle with a base of $5\sqrt{2}$ cm and a perpendicular height of $3\sqrt{18}$ cm.

4 A rectangular field is $26\sqrt{2}$ m long and it has an area of $390\sqrt{6}$ m^2.
Find the width of the field.

5 Write the following in their simplest rationalised form.

a $\frac{1}{2+\sqrt{3}}$

b $\frac{2}{5-\sqrt{2}}$

c $\frac{7}{8-\sqrt{5}}$

d $\frac{5}{11+3\sqrt{2}}$

e $\frac{\sqrt{5}}{13+2\sqrt{7}}$

f $\frac{1}{\sqrt{3}+\sqrt{2}}$

g $\frac{\sqrt{5}}{\sqrt{10}-\sqrt{7}}$

6 Given that $\frac{1}{3+\sqrt{5}} = \frac{3-\sqrt{5}}{a}$, find the value of a.

7 Given that $\frac{-5+\sqrt{3}}{b} = \frac{2}{-5-\sqrt{3}}$, find the value of b.

8 Given that $\frac{\sqrt{19}-\sqrt{3}}{c} = \frac{c}{\sqrt{19}+\sqrt{3}}$, find the possible values of c.

9 Given that $\frac{2d}{3\sqrt{17}-2\sqrt{7}} = \frac{3\sqrt{17}+2\sqrt{7}}{5d}$, find the possible values of d.
Give your answers in rationalised surd form.

10 Find the value of x, given that $\frac{x}{15+5\sqrt{5}} + \frac{x}{15-5\sqrt{5}} = 3$

Discussion activity 1.5

1 b The answers (7, 22, 37 and 4) are all rational numbers

c $a = 7, b = \sqrt{2}$

d $a = 9, b = \sqrt{3}$

e $a = \sqrt{7}, b = 11$

f $a = \sqrt{12} = 2\sqrt{3},\ b = \sqrt{18} = 3\sqrt{2}$

Exercise 1.15B

1 a 21

b 28

c 9

d $\frac{5}{24}$

2 a $4\sqrt{2}$

b $7\sqrt{3}$

c $8\sqrt{5}$

d $\frac{3\sqrt{2}}{2}$

e $\frac{51\sqrt{10}}{2}$

f $\frac{5\sqrt{3}}{3}$

g $\frac{16\sqrt{3}}{5}$

3 a $245\,\text{cm}^2$

b $160\,\text{cm}^2$

c $45\,\text{cm}^2$

4 $15\sqrt{3}$ m

5 a $2-\sqrt{3}$

b $\frac{10+2\sqrt{2}}{23}$

c $\frac{56+7\sqrt{5}}{59}$

d $\frac{55-15\sqrt{2}}{103}$

e $\frac{13\sqrt{5}-2\sqrt{35}}{141}$

f $\sqrt{3}-\sqrt{2}$

g $\frac{5\sqrt{2}+\sqrt{35}}{3}$

6 $a = 4$

7 $b = 11$

8 $c = -4$ or $c = 4$

9 $d = -\frac{5\sqrt{2}}{2}$ or $d = \frac{5\sqrt{2}}{2}$

10 $x = 10$

Discussion activity 1.6

1 a i $(\tan 60°)^2 = 3$

ii $(\sin 60°)^2 = 0.75 = \frac{3}{4}$

iii $(-1+4\sin 54°)^2 = 5$

b $\tan 60° = \sqrt{3};\ \sin 60° = \frac{\sqrt{3}}{2};\ \sin 54° = \frac{1+\sqrt{5}}{4}$

c $p = 8$

d $\sin 36° = \sqrt{\frac{5-\sqrt{5}}{8}}$

End-of-chapter 1 review exercise

1 a $0.0\dot{1}\dot{3}$

b $\frac{14}{33}$

c 2

2 a 63.5 mm

b 200 mm

3 a $7.5\,\text{cm}^2$

b 3 cm

4 a 49.7 km/h or 49.72 to 49.74 km/h

b 45.6 km/h

5 a 16%

b No; maximum possible mass is 23.25 kg

c \$72

d \$36.44

e

7 nights at \$**115** per night	\$**131**
4 dinners at \$**32.75** each	\$**936**

6 a $\frac{23}{24}$ and $\frac{27}{28}$

b $k = 300$

c $\frac{4n-1}{4n}$